CONTROL OF PARTIALLY SPECIFIED BOOLEAN NETWORKS

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Eva Šmijáková, *Control of Partially Specified Boolean Networks* © 2026

DECLARATION

Hereby I declare that this thesis is my original authorial work, which I have worked out on my own. All sources, references, and literature used or excerpted during the elaboration of this work are properly cited and listed in completed reference to the due source. During the preparation of this thesis, I used the following AI tools to improve the clarity and style of my writing:

- Grammarly
- ChatGPT
- GitHub Copilot

Additionally, I used the following tool to assist with software development:

- GitHub Copilot

I declare that I used these tools in accordance with the principles of academic integrity. I checked the content and took full responsibility for it.

Eva Šmijáková

ABSTRACT

In various fields, an attractive challenge is to influence the systems to align with desired behaviors. Formal methods offer ways to address this challenge by utilizing abstractions of real-world systems to compute the solutions. This thesis focuses on the control of biological systems, which contributes to solving cell reprogramming – a significant challenge in regenerative medicine.

Biological systems are modeled using a variety of approaches, including Boolean networks (BNs), which thanks to their simplicity and expressiveness effectively capture interactions at the cellular level. While control problems in the context of fully specified BNs have been extensively studied, for cases when Boolean functions are not fully known, the control problem has not yet received significant attention.

Partially specified Boolean networks (PSBNs) accommodate uncertainties in the model by allowing function symbols. However, this introduces complexities in the control problem, including variations in the state transition graph between varying interpretations and optimization challenges in minimizing perturbations while ensuring target reachability.

This thesis deals with these challenges and brings the control problem to the partially specified setting of Boolean networks. The state perturbations are used to exercise control on the models since their implementation in practice can be realized via gene knockouts or overexpressions. The most significant difference of PSBN control is, that instead of minimal perturbation a large set of possible perturbations is computed along with the perturbation robustness to the uncertainties posed by the model.

We present two flavors of the PSBN control problem – source-target control and phenotype control. Two methods were developed to deal with the source-target problem – a semi-symbolic parallel method, which uses just one-step perturbations and a symbolic method, which is able to employ one-step, temporary and permanent perturbations. The weakness of source-target control lies in the necessity to know the target attractor beforehand, which can be highly inconvenient for PSBN. This weakness is mitigated by the phenotype control, where only characteristic traits of the target are sufficient to be specified and also, the control works globally regardless of the source state. A semi-symbolic method of permanent perturbations was developed to solve this problem. Finally, software encompassing all these methods is described and their applicability is presented in real-world case studies.

PUBLICATIONS

This manuscript encompasses and expands the results from the previously published works. Here, I describe their relationship to this manuscript. The detailed structure of this work is then summarized in Chapter 1. The groundwork for concepts that are probed in this manuscript was laid in my Ph.D. thesis proposal [Šm21].

My first publication [CMSB'20] is based on the results that I have achieved in my master thesis. Since all consequent work is done on formal methods applied to systems biology, this publication gave me a solid foundation in the field.

Then I focused on the topic of control of partially specified Boolean networks, which is the backbone of this manuscript. The two published works [MATHEM'21, BIOSYS'23] focus on source-target control and their results are demonstrated in **??**.

The findings from these works inspired me to investigate phenotype control of partially specified Boolean networks because it alleviates the weaknesses of source-target control in the partially specified setting. The results achieved with this method were published in [CMSB'23] and here are described in **??**.

The methods I have developed in [BIOSYS'23, CMSB'23] were then included in the AEON.py tool. This tool was presented as the Application note [BIOINF'23]. Later, the control methods were incorporated into the frontend of the AEON tool as is presented in [CMSB'25]. The relevant parts of the tools are described here in Chapter 2.

Finally, I have proposed a novel framework to design control experiments for refinement of the partially specified Boolean networks. The framework and relevant case studies are also demonstrated in the journal article [SUBMITTED]. This framework along with other case studies for control of partially specified Boolean networks, are summarized in **??**.

The following list chronologically enumerates the publications I co-authored. For each publication, I describe my involvement in the manuscript preparation, and an approximate percentage indicating my contribution.

[SUBMITTED] *Control-Guided Refinement of Partially Specified Boolean Networks: Applications to RTK Signalling* (E. Šmijáková, L. Brim, S. Pastva, D. Šafránek) TODO cite

> *A journal original paper. I have conceptualized the novel framework for control-guided refinement of partially specified Boolean networks as well as methods for controlling partially specified networks towards oscillatory phenotypes. The conducted experiments demonstrating the*

proposed methods as well as wrote the major part of the paper. (70%)

[CMSB'25] *AEON 2025: Robust Control of Partially-Specified Boolean Networks* (V. Veselý, E. Šmijáková, S. Pastva, N. Beneš, D. Šafránek) [Ves+25]

A tool conference paper. The main author developed extension of the AEON tool frontend based on the methods I have previously developed. I have contributed to the implementation and testing of the tool and written a significant part of the paper. (30%)

[CMSB'23] *Phenotype Control of Partially Specified Boolean Networks* (L. Brim, S. Pastva, D. Šafránek, E. Šmijáková) [Ben+23b]

A regular conference paper. I have conceptualized the novel type of control problem with better usability than the previous ones. I have proposed and implemented the methods. I have proposed and realized case studies that demonstrated the applicability of the developed methods. I did a significant part of the writing. (70%)

[BIOSYS'23] *Temporary and Permanent Control of Partially Specified Boolean Networks* (L. Brim, S. Pastva, D. Šafránek, E. Šmijáková) [Bri+23]

A full-length journal article. I proposed, developed, and implemented the novel methods introduced in the paper. I designed and executed the experiments demonstrating the capabilities of the methods and did an extensive part of the writing. (70%)

[BIOINF'23] *AEON.py: Python library for attractor analysis in asynchronous Boolean networks* (N. Beneš, L. Brim, O. Huvar, S. Pastva, D. Šafránek, E. Šmijáková) [Ben+22a]

An application note published in the journal. I was the author of the control methods included in the published tool paper. I have described the technical details of these methods. (30%)

[MATHEM'20] *Parallel One-Step Control of Parametrised Networks* (L. Brim, S. Pastva, D. Šafránek, E. Šmijáková) [Bri+21]

An article published in a journal's special issue on Boolean Networks Models in Science and Engineering. I have conceptualized the novel problem of the control of partially specified Boolean networks. I have proposed the algorithm that solves this problem and collaborated on its implementation as well as written a major part of the manuscript. (70%)

[CMSB'20] *Parallel Parameter Synthesis for Multi-affine Hybrid Systems from Hybrid CTL Specifications* (E. Šmijáková, S. Pastva, D. Šafránek, L. Brim) [Šmi+20]

A regular conference paper. The article presents the results achieved in my master's thesis [Šm19]. I have proposed and implemented a method based on a previous work of the SyBiLa research group. I significantly participated in the manuscript writing. (60%)

ACKNOWLEDGEMENTS

As no man is an island, no thesis is completed alone. I would like to express my sincere gratitude to my supervisor, Luboš Brim, for giving me the opportunity to work on this topic and for his continuous support and guidance throughout my PhD. I am equally grateful to David Šafránek for taking up the baton and for his help and support during my studies.

I would also like to thank all members of the SyBiLa group for their collaboration and for keeping me inspired. Special thanks go to Samuel Pastva and Matej Troják for sharing the hardships of doctoral studies with me and for their mutual support, to Vojtěch Veselý for development of the AEON tool user interface.

My deepest thanks go to my significant other, Michal, who has been my inspiration trough out of the whole journey. Enrolling in the PhD program at the same time and reaching the finish line sooner than I did, he has been a constant source of motivation. I am grateful for his understanding and support during the most challenging times.

Finally, I would like to thank my family for their unwavering support and encouragement throughout my studies, and my friends and colleagues for their understanding, support, and for resisting the urge to ask about the progress of my thesis.

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1 INTRODUCTION

An attractive challenge for many scientific fields is finding ways to influence systems, so they behave as desired. In reality, this can be very difficult as there might be many different ways to interfere with the systems and at the same time, some interventions can be more efficient than others. Thanks to formal methods such as control or parameter synthesis we can use abstractions of the systems (models) and compute the solution to the original real-world problem.

The focus of this work is on the control of biological systems. Therefore, the problem falls into the scope of the *systems biology* field where the aim is to apply formal methods developed by computer science to the biological models. The current focus of systems biology is on the cellular level [Car05] as it is the basis of biological systems and even this basic level is still considered difficult to model as there might be hundreds of interactions in the play. The task of controlling cells is called *cell reprogramming*, and it is one of the most significant challenges of regenerative medicine [CD12].

Systems biology uses a variety of models ranging from discrete (e.g. Boolean networks [Sch+20]), through continuous (e.g. ODE [GIP13]), to hybrid models [Gro+93]. A very efficient model to capture numerous interactions at the cellular level is *Boolean network* (BN) because this model is both simple and expressive. Moreover, BNs have applications in many other areas, including circuit theory and behavioral science [Rol+15; KS14].

BNs are composed of two parts: a finite set of Boolean *variables* representing genes or other biochemical substances and *update functions* which specify the way variables dynamically change their value based on influences from other variables. BNs generate a finite discrete state-space which makes them convenient for computational analyses. When starting from any initial state, the BN eventually establishes in some single state or a set of states. These components of BN state space are called *attractors*. This notion can be viewed as an abstraction for phenotypes of the modeled biological systems.

A computational model is considered controllable if we can assure that from some initial state, it reaches a desired final state in a finite amount of steps. When controlling BNs, we are interested in BN reaching the desired attractor. This property is well-studied in the context of BNs without unknown parts [ZA15; Man+19; SP20; CFTS20]. However, the problem has not received a lot of attention in the partially specified setting yet.

The definition of control is rather broad and allows many variations of the problem for BNs. To change the normal behavior of BN, understandably, we need to interfere with the model in some way. We call these interventions *perturbations*. However, we can consider many types of perturbations. They vary in three aspects. Firstly, we can either fix variable values to some fixed value or change the rules of their evolution. Secondly, we can apply the perturbation just as a single transition step, or we can hold the perturbations for more transition steps, even permanent perturbations are considered. Lastly, the perturbation can be applied only once, or we consider applying perturbations and releasing perturbations at any time.

Apart from different kinds of perturbations, control problems also vary in terms of what we are trying to achieve. For example, two elementary types of control problems are: (i) achieving a single desired target attractor irrespective of the current state (target control) [KPC13], (ii) achieving control between every pair of attractors (full-control) [Fie+13], and (iii) reaching the set of states satisfying the given conditions (phenotype control) [CFTS20].

In practice, the exact Boolean functions of the model may not be precisely known due to various uncertainties such as insufficient experimental knowledge [Gri+13], inconsistent observations [GGC16], genetic mutations [Mar+16], or any other ambiguities. In contrast, there is typically good evidence of the fact that a variable regulates another one. This issue can be addressed by allowing BNs to contain function symbols. *Partially specified Boolean networks* (PSBNs) [Zou13; Ben+19] allows to specify only regulators of a variable and thus capture multiple variants of the possible actual behavior of variables without conducting many more expensive experimental observations. Every interpretation of the unknown parts can be understood as an instance of a standard BN. The number of these instances can be doubly exponential in the worst case [WSA12].

The partially specified setting brings many new challenges to the control problem. First, the nature of BN state transition graph may change dramatically between the interpretations [Ben+19; Ben+20]. Even the original goal of the standard control – to stabilize in a target attractor – might not be feasible in some instances as the given attractor is not present in them. Secondly, the BN control problem is optimizing – we are trying to minimize the number of perturbations done to a network, while we must guarantee that the target attractor is reached in all BN non-deterministic runs. It was shown that for many biological models, controlling values of only a few variables is sufficient to drive the network into any attractor of the BN [ZA15]. On the contrary, this is usually not the case for the PSBNs. Due to the uncertainties, we would often need to fix the value of many variables to guarantee PSBN to reach the given attractor in all network instances. Such an approach is not applicable in practice as each perturbation

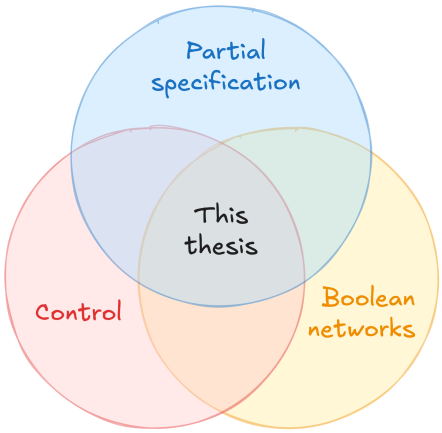

Figure 1: The confluence of concepts studied in this thesis.

can be very costly in the real environment. Therefore, the optimization criteria should be adjusted.

In our current research line, we decided to treat the interpretations solely as unknown knowledge and consider only the state perturbations that fix the value of variables. To cope with minimization criteria, the output of the control problem for PSBN is a complete *mapping* from the domain of perturbations to the domain of relevant interpretations. This way, it is possible to manually explore the available perturbations and consider both the cost of perturbation and the *robustness* of perturbation – the proportion of interpretations for which the perturbation ensures PSBN stabilization in the given attractor with all considered interpretations.

CONTRIBUTION

The contribution of this thesis lies in its integration of three distinct concepts: control, Boolean network and systems with partial specification. Such a confluence has not yet been explored previously. The Boolean networks are a well-studied model of system biology. Also, control is a commonly employed formal method in systems biology and the topic of controlling BNs was already thoroughly studied as we will see in **??**. On the other hand, the promising model of partially specified Boolean networks is just beginning its rise. Other formal methods such as attractor search [Ben+21], bifurcation [Ben+22b; Pas22], or parameter synthesis [Ben+23a] are also being investigated only relatively recently.

The partial specification of BNs thus makes this work truly novel, although the specific approaches are often inspired by previously investigated intersections of the topics as we will see in the thesis. The objectives of this thesis are four-fold:

1. Formally define control problem for partially specified Boolean networks
2. Develop efficient methods that solve the control problem for partially specified Boolean networks
3. Bundle the developed methods to a software toolkit
4. Demonstrate applicability for biological systems

Now let us have a look at these goals in detail and describe in what parts of this thesis they are fulfilled.

GOAL 1 – PROBLEM DEFINITION. Problem definition is an essential contribution of this work since the problem of PSBN control has not been formalized prior to my work. Several crucial aspects need to be considered for the problem definition. Firstly, attractors of the PSBN vary according to the interpretation. Therefore, searching for non-permanent perturbations that drive the network into a target attractor might not make sense for interpretations, where the given attractor is not present in the first place. Secondly, the practical realization of perturbations requires non-trivial effort; therefore, in a normal BN setting the number of perturbations is sought to be minimal. However, there is rarely a solution with a few perturbations in the partially specified setting, which would work for a big portion of interpretations. Thus, optimization criteria should be readjusted accordingly.

The achievement of this goal is presented in **??** where we define *source-target* control for PSBN which result is a relation of perturbations with interpretations for which they work as opposed to standard control goal where only minimal perturbation sets are returned. However, the weakness of the source-target control is, that an exact target attractor might not be known in advance, since the landscape of attractors may change between interpretations [Ben+22b]. This caveat is resolved by a more general notion of control – *phenotype* control which is defined in **??** where only target traits are necessary to be known in advance.

GOAL 2 – EFFICIENT METHODS DEVELOPMENT. As described before, the control problem for fully specified asynchronous BNs has been well-studied already. Therefore, after control for PSBN is formally defined, in theory, we could use the existing methods for asynchronous BN control to solve the problem for every interpretation distinctly. However, the number of these interpretations can be doubly exponential, meaning even a very efficient method for BN control would need to be run too many times to be handled in a realistic time. That is why seeking a novel, efficient approach for the partially specified setting is crucial.

Three types of methods were developed to solve the posed variants of the problem.

1. Semi-symbolic parallel one-step source-target control method for PSBN which is described in **??** [Bri+21]
2. Symbolic one-step, temporary, and permanent source-target control for PSBN which are described in **??** [Bri+23]
3. Semi-symbolic permanent phenotype control method for PSBN which is described in **??** [Ben+23b]

The aforementioned sections present algorithms, and simple (i.e., quantitative) demonstrations of them.

GOAL 3 – SOFTWARE TOOLKIT FOR METHODS. In order to make the developed methods publicly available, they have become a part of the BioDivine toolkit [Bar+09]. A library in Rust based on AEON tool [Ben+20] was developed in order to make methods fast and performant. Later, the library was wrapped into Python language as part of AEON.Py tool [Ben+22a]. This helps to make the methods more convenient to work with for the wider scientific audience because the language is easier to learn and offers popular notebook environments such as JuPyter [Bar21]. The Python version also has become part of the CoLoMoTo Interactive Notebook tools [Nal+15; Lev+18]. The software technical specification and solved challenges are in detail described in Chapter 2.

GOAL 4 – APPLICABILITY FOR BIOLOGICAL SYSTEMS. Last, but not least, since the topic of Boolean network control is tightly tied to the cell reprogramming problem of regenerative medicine, it is very suitable to demonstrate that the developed methods are serving the original motivation goal. Moreover, such case studies can serve as a tutorial for the developed software tool. All case studies are presented in **??**.

THESIS STRUCTURE

TODO

2 SOFTWARE

In this chapter, we describe the technical details of the software tools we developed for the control of Boolean networks. We first give an overview of the software. Then we describe the implementation underlying backend details of the main components of the software. Finally, we describe the python package and the web application that we developed to make the software accessible to a wider audience.

2.1 SOFTWARE OVERVIEW

The software for the control of PSBN networks was developed as part of the BioDivine tools[1] developed by the SyBiLa (Systems Biology Laboratory) research group at Masaryk University [2].

The tools were initially oriented on a "colored model checking" of parametrized continuous models of biological systems [Bri+15]. The approach was also later extended to hybrid models [Šmi+20]. The main idea behind colored model checking was encoding the continuous models as discrete color-labelled transition systems, and then applying SAT model checking techniques to analyze their behavior. This approach also enabled solving the parameter synthesis and attractor bifurcation problems for these models [Ben+16; Ben+17].

Building on top of the previous work, the group shifted focus from discretized continuous models to Boolean networks, which are a more direct "natively" discrete modeling formalism. The former parameters of continuous models and their paramters synthesis was reoriented from continuous values of parameters to unknown functions of the network structure.

At first, the tools were mostly focused on the bifurcation analysis of PSBN [Ben+19]. To nurture the adoption of the tool, a web application was developed to make the tool accessible to a wider audience [Ben+20]. The web application allowed users to upload their own partially specified Boolean network models, adjust them and run the bifurcation analysis.

Since globally there was not a lot of tools supporting analysis of PSBN models, the Aeon tools were extended to support also other types of analysis. In particular, we implemented the control of PSBN models, which is the main topic of this thesis. The control of PSBN models was implemented as a new component of the software, and it was integrated into the web application as well. Later, to make the

1 https://sybila.fi.muni.cz/tools.html
2 https://sybila.fi.muni.cz/index.html

control of PSBN models more accessible to users who prefer working with code, we also implemented a python package that provides an interface to the control algorithms implemented in the software [Ben+22a]. The tool also provides all underlying algorithms and data structures such as the symbolic representation of the state space, and the algorithms for computing the attractors and their basins of attraction. The tool is also being extended by higher-level algorithms for the analysis of PSBN models, such model checking of temporal properties, and the analysis of the structure of the state space [Ben+23a; Ben+24].

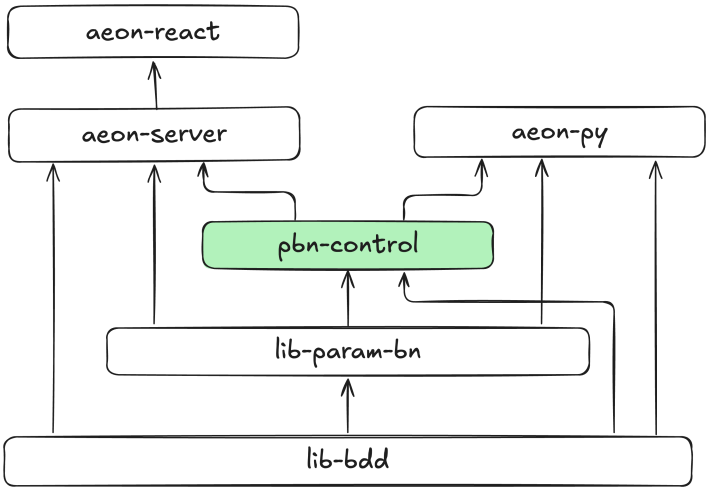

Figure 2: **Biodivine Aeon software architecture overview.** The figure depicts components of the biodivine software related to the control of Boolean networks. The green box represents the main Rust library, which contains the core algorithms and data structures for Boolean network control. The arrows indicate the dependencies between the components.

In Figure 2, we give an overview of the software architecture of the Aeon tools. The core components are implemented in Rust language because of its performance, memory safety and mature developer experience. Component lib-bdd is the low-level library for the symbolic representation of the state space using binary decision diagrams (BDDs). It is independent of Boolean networks and can be used for any application that requires symbolic representation of state spaces.

Component lib-pbn is the main library for the performant analysis of PSBN models, which contains the algorithms for computing attractors, reachability algorithms and bifurcation. On top of the two libraries, we implemented the control of PSBN models as a new component of the software. Since the control of PSBN requires custom encoding of the perturbations it also relies on the lib-bdd library for the symbolic representation of the state space. The control component

also relies on the lib-pbn library for computing the attractors and their basins of attraction, which are used in the control algorithms.

Finally, we implemented a web application that is built on top of all the rust components. Aeon-server provides a server functionality such as validation of the uploaded models and progress reporting to interact with the software through a web interface. The web application (aeon-react) than uses just the server component of the software, which provides an API for the users to interact with the software through the web interface. Later, also a python package aeon-py was implemented to provide a convenient python interface for all low-level algorithms and data structures.

2.2 IMPLEMENTATION DETAILS

2.3 AEON WEB APPLICATION

Aeon web application is a web-based interface for the analysis of PSBN models. It allows users to upload their own models, adjust them and run various analyses, including the control of PSBN models. The initial version of the web application was developed to make the bifurcation analysis of PSBN models accessible to a wider audience. The initial version was first released in 2020 [Ben+20].

In version 0.5.0 of the web application, we integrated the control of PSBN models as a new component of the software [Ves+25]. During implementation of the new control module, it was discovered, that technical debt in the codebase of the web application was making it difficult to implement the new control module. Therefore, it was decided to refactor the web application to make it more modular and maintainable. This has led to a major refactor of the web application, which was released in version 0.6.0 and which is also briefly presented here.

AEON tool is available at `https://biodivine.fi.muni.cz/aeon/` as an online tool. Due to the potential high computational complexity of the analyses, SyBiLa group does not provide a public instance of the web application server, but instead provides a downloadable version of the server that users can run on their own machines.

The main flow of how a model can be uploaded and analyzed in the web application is shown in Figure 3. The user first needs to download and run the server component of the software on their own machine. Then, they can access the web application functionality through a web browser. The user can upload their own PSBN model in the form of a text file, which is then validated by the server or select one of the example models provided by the web application. Once the model is loaded, the user can adjust the model by changing the model functions, or by changing the regulations of the model.

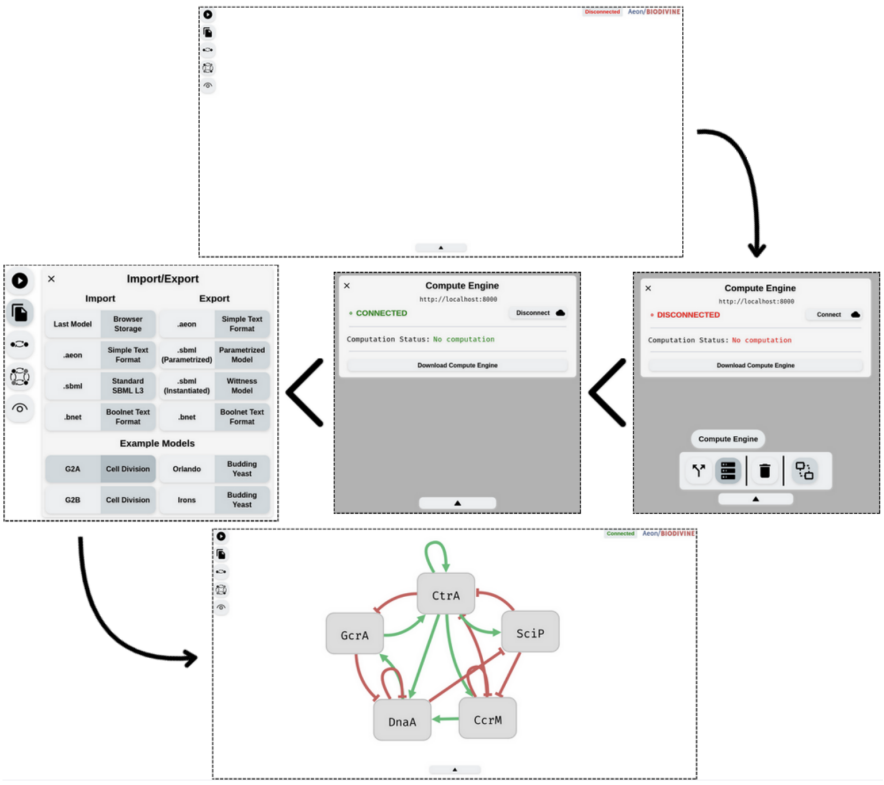

Figure 3: **Biodivine Aeon web application interface.** From [**Vesely2026**]. The figure shows the main interface of the Aeon web application, where users can upload and load their own PSBN.

After the model is loaded, the user can run various analyses on the model, including the permanent phenotype control of PSBN models. The source-target control is not support via Aeon UI. It could be implemented later if there is a user interest. The control interface of the web application is shown in Figure 4. The user can select variables that can be perturbed, the type of the phenotype control (allowed, required or forbidden oscillation) and the target phenotype that they want to reach. The user can also adjust the parameters of the control algorithm, such as the minimal required robustness, maximum number of perturbations allowed or maximum number of results.

After the control algorithm is run, the results are displayed in the web application, where the user can inspect the results and download in a CSV format for further analysis. The results include the comprehensive statistics about the control problem, such as elapsed time, number of interpretations that can be controlled, the number of perturbations found, highest robustness achieved, and other relevant information. Moreover, a full list of all possible perturbations that can be applied to reach the target phenotype is also available. The list can be filtered and sorted by various criteria, such as the number of perturbations, the robustness of the perturbation, or the variables that are perturbed.

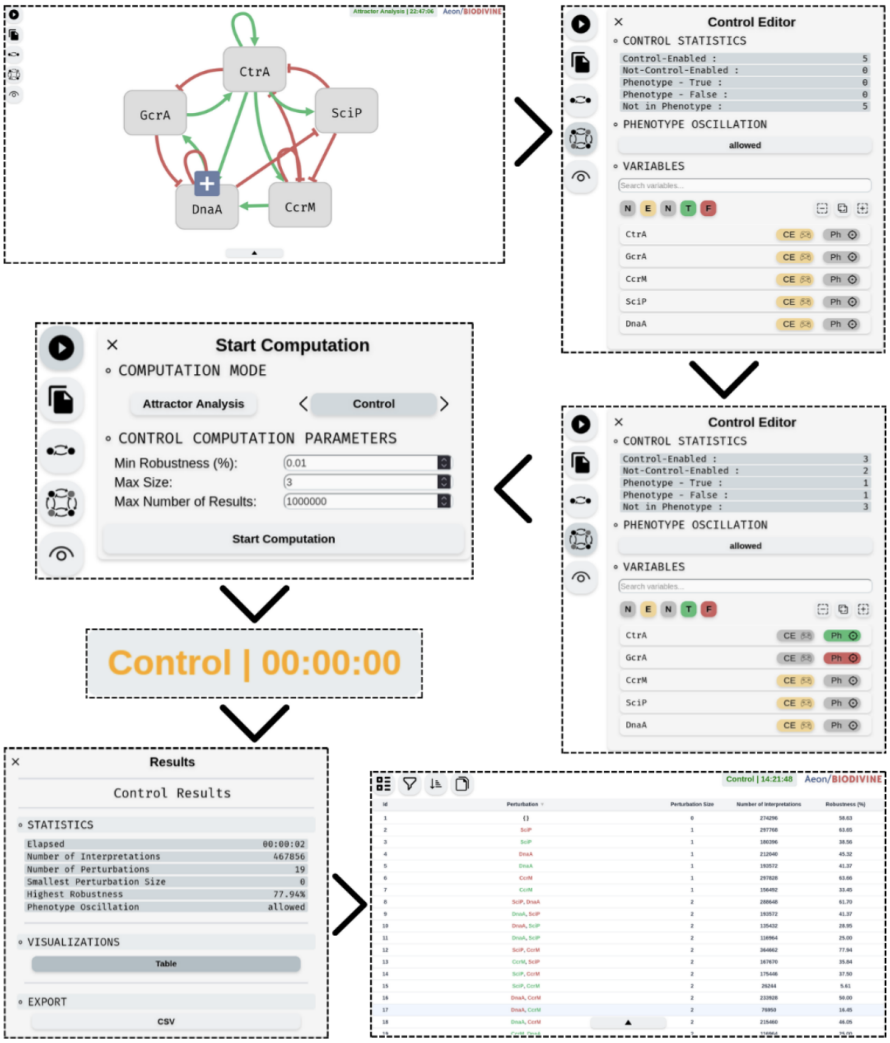

Figure 4: **Biodivine Aeon web application control interface.** From [**Vesely2026**]. The figure shows the control interface of the Aeon web application, where users can control their own PSBN.

2.4 AEON.PY

Since it can be difficult to develop a web application that covers all the functionality of the a very technical software, we also implemented a python package that provides an interface to the control algorithms implemented in the software [Ben+22a]. The python package is available on PyPI and as part of the CoLoMoTo docker image [Nal+15].

The CoLoMoTo toolset provides a robust and reproducible computational framework for the analysis of Boolean networks, enabling both qualitative and formal investigation of complex biological regulatory systems. Its Jupyter environment based on Docker ensures interoperability and reproducibility while facilitating scalable explo-

ration of state-space dynamics under different updating schemes. Since it is widely used in the community of Boolean network modeling, we decided to make the Aeon control algorithms available as part of the CoLoMoTo docker image. This allows users who are already using the CoLoMoTo toolset to easily integrate the control algorithms into their existing workflows.

The python package provides a convenient interface for users who prefer working with code rather than a web application. It allows users to load their own PSBN models, adjust them and run the control algorithms directly from their python and run a large scale experiments programatically.

CONCLUSIONS

Conclude here...

A

NOTATION

	Boolean algebra related symbols
b	Boolean value (true or false)
$\mathbb{B}$	Set of Boolean values (true or false)
$*$	Undefined Boolean value (might be true or false)
f	A function
$\mathbb{B}_*$	Set of Boolean values including the undefined value
$x[i \mapsto b]$	A vector x with the i-th element replaced by b
	Networks - common
n	Size of a network (number of its nodes – variables)
$\mathcal{G}$	Regulatory network
$\mathbb{V}$	Vertices of a network
$\mathbb{E}$	Edges of a network
S	Subspace of a network
$\mathcal{N}$	Boolean network instance
STG	Asynchronous state transition graph
π	A run (trajectory) of a (partially specified) Boolean network
$\mathcal{E}(\texttt{var1, var2})$	Influence of `var1` on `var2` is observable in STG
`var`	A font for variable names
$\mathcal{A}$	An attractor of a (partially specified) Boolean network
$\mathbb{A}$	Set of all attractors of a (partially specified) Boolean network
$\mathcal{U}$	An observed character of a network
$\mathcal{T}$	A trait of a network
ϕ	A phenotype of a network
	Partially specified Boolean networks
$\mathcal{M}$	Partially specified Boolean network
g	A function symbol of a partially specified Boolean network
$\mathbb{G}$	Set of all function symbols of a partially specified Boolean network
E	An expression of a partially specified Boolean network
I	An interpretation of a partially specified Boolean network
$\mathcal{I}$	A set of interpretations
$\mathbb{I}$	Set of all interpretations of a partially specified Boolean network

	Control
Q	A perturbation of a network
$\mathbb{Q}$	A set of all admissible perturbations
ρ	Robustness of a perturbation towards considered set of interpretations

B

DIGITAL ATTACHMENTS

Lorem Ipsum..

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